Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
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Practice Exam — Week 10: Forecasting

Duration: 60 minutes · No calculator · Answer on paper

Name: ______________________ Surname: ______________________

SCIPER (Student Card Number): ______________________

Exercise	Points
1	
2	
3	
4	
Total	

Justify every answer. Throughout, ε_t is a mean-zero white noise of variance σ^2.

Problem 1 (5 points)

(Forecasting a causal AR(1): point forecast, error variance, interval.) Let $\{X_t\}$ be the zero-mean Gaussian AR(1) process

$$X_t = \phi X_{t-1} + \varepsilon_t, \qquad |\phi| < 1, \quad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2),$$

and write X_{n+h}^n for the best linear predictor of X_{n+h} based on $X_1, \ldots, X_n$, with h-step forecast error $e_n(h) = X_{n+h} - X_{n+h}^n$.

(a) Give the MA(∞) weights $\{\psi_j\}$ of the process. Using that the process is Gaussian (so the best linear predictor equals the conditional expectation), show that the point forecast is $X_{n+h}^n = \phi^h X_n$, and state its limit as $h \to \infty$. (1.5 pts)

(b) Show that the h-step forecast-error variance is

$$\operatorname{Var}\big(e_n(h)\big) = \sigma^2 \, \frac{1 - \phi^{2h}}{1 - \phi^2},$$

and verify that $\operatorname{Var}(e_n(h)) \to \gamma_0$ (the marginal variance) as $h \to \infty$. (2 pts)

(c) Now fix $\phi = 0.6$ and $\sigma^2 = 4$. Report X_{n+2}^n and the numerical forecast-error variances $\operatorname{Var}(e_n(1)), \operatorname{Var}(e_n(2))$, then give the 95% Gaussian prediction interval for X_{n+2} (use $z_{0.975} = 1.96$). (1.5 pts)

Problem 2 (5 points)

(The prediction equations, and one-step AR(2) forecasts.) Let $\{X_t\}$ be a zero-mean second-order stationary process with autocovariance $\{\gamma_\tau\}$, and let

$$X_{n+h}^n = \sum_{j=1}^{n} \beta_j X_j$$

be its best linear predictor of X_{n+h} based on $X = (X_1, \ldots, X_n)^\mathsf{T}$.

(a) Starting from the prediction mean square error $S(\beta) = \mathbb{E}\big[(X_{n+h} - \beta^\mathsf{T} X)^2\big]$, derive the *prediction equations* $\Gamma_n \beta = \gamma_{[h]}$, where $\Gamma_n = [\gamma_{i-j}]_{i,j=1}^n$ and $\gamma_{[h]} = (\gamma_{n+h-1}, \ldots, \gamma_h)^\mathsf{T}$, and show that this is equivalent to the orthogonality condition $\mathbb{E}[(X_{n+h} - X_{n+h}^n)X_k] = 0$ for $k = 1, \ldots, n$. (2 pts)

(b) Assuming Γ_n invertible, deduce the closed forms

$$X_{n+h}^n = \gamma_{[h]}^\mathsf{T} \Gamma_n^{-1} X, \qquad P_{n+h}^n = \gamma_0 - \gamma_{[h]}^\mathsf{T} \Gamma_n^{-1} \gamma_{[h]}. (1.5 \text{ pts})$$

(c) Let $\{X_t\}$ be the causal AR(2) process $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$. Use the result of part (a) to show that the best one-step predictors based on X_1 and on X_1, X_2 are

$$X_2^1 = \frac{\gamma_1}{\gamma_0} X_1 = \rho_1 X_1, \qquad X_3^2 = \phi_1 X_2 + \phi_2 X_1.$$

(For the second, you need *not* invert Γ_2.) (1.5 pts)

Problem 3 (5 points)

(The causal-ARMA predictor via ψ-weights, and a worked ARMA(1,1).)

(a) For a causal ARMA process with linear representation $X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$, prove that the best linear predictor of X_{n+h} based on $X_1, \ldots, X_n$ is

$$X_{n+h}^n = \sum_{j=h}^{\infty} \psi_j \, \varepsilon_{n+h-j}, \qquad \text{with} \qquad P_{n+h}^n = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2.$$

(*Hint:* any linear predictor based on $X_1, \ldots, X_n$ can be written $\sum_{j \geq 0} c_j \varepsilon_{n-j}$; expand the MSE and minimise over the c_j.) (2 pts)

(b) Consider the causal, invertible ARMA(1,1) process

$$X_t = \tfrac{1}{2} X_{t-1} + \varepsilon_t + \tfrac{3}{10} \varepsilon_{t-1}, \qquad \sigma^2 = 2.$$

Compute the first three ψ-weights ψ_0, ψ_1, ψ_2 (from $\Theta(B)/\Phi(B)$), and give the point forecasts X_{n+1}^n and X_{n+2}^n in terms of X_n and the residual $\hat{\varepsilon}_n$. (1.5 pts)

(c) Report the forecast-error variances $P_{n+1}^n, P_{n+2}^n, P_{n+3}^n$ as exact numbers, and explain in one sentence the value γ_0 they approach as $h \to \infty$ (you may use $\gamma_0 = \sigma^2(1 + 2\phi\theta + \theta^2)/(1 - \phi^2)$ for an ARMA(1,1)). (1.5 pts)

Problem 4 (5 points)

(Revision of Week 9: VAR stability and the diagonal VAR(1) covariance.)

(a) Consider the bivariate VAR(2) process $X_t = \Phi_1 X_{t-1} + \Phi_2 X_{t-2} + \varepsilon_t$ with

$$\Phi_1 = \begin{pmatrix} 0.4 & 0 \\ 0.5 & 0.9 \end{pmatrix}, \qquad \Phi_2 = \begin{pmatrix} 0 & 0 \\ 0.3 & -0.2 \end{pmatrix}.$$

Form the reverse characteristic polynomial $\det\big(I_2 - \Phi_1 z - \Phi_2 z^2\big)$, exploit the lower-triangular structure to factor it, find all its roots, and decide whether the VAR(2) is stable. State the companion-matrix (F) form of the stability test and explain why it gives the same conclusion. (3 pts)

(b) Now take the *diagonal* VAR(1) process $X_t = \Phi\, X_{t-1} + \varepsilon_t$ with

$$\Phi = \operatorname{diag}\big(\tfrac{1}{2}, \tfrac{1}{4}\big), \qquad \{\varepsilon_t\} \sim \mathrm{WN}(0, I_2).$$

Write its MA(∞) representation and compute the stationary covariance Γ_0 and the lag-1 matrix Γ_1 in closed form (exact fractions). (1.5 pts)

(c) For the process of part (b), what is the coherence $r_{1,2}(f)$ between the two channels at every frequency, and why? (0.5 pts)